

UPI Adoption vs. Fraud Risk: A Trend Analysis

Analyzing India's digital payments growth against fraud trends using official NPCI & RBI data (FY2016-17 – FY2025-26)

787.5B

Total UPI transactions processed (2016–26)

Rs. 1,136L Cr

Total transaction value processed (2016–26)

72.7%

Decline in fraud ratio since FY18-19 peak

6.6%

Forecast model error (MAPE, 6-month test)

Headline Finding

UPI's Card/Internet fraud-to-transaction-value ratio peaked at 0.0086% in FY2018-19 and has declined ~73% to 0.0024% by FY2024-25 — despite transaction value growing over 300x in the same period. This suggests RBI/NPCI security interventions have meaningfully outpaced the growth in fraud activity, even as digital payment adoption has scaled nationally.

Methodology

Data & Tools

Data sources: NPCI UPI Product Statistics (monthly volume/value, Apr 2016 – Aug 2026); RBI Report on Trend and Progress of Banking in India 2024-25, Appendix Table IV.7 (Card/Internet fraud, annual).

Pipeline: Python (pandas) for cleaning & merging multi-source data → SQL (PostgreSQL, window functions) for YoY growth & fraud-ratio aggregation → Power BI for the interactive dashboard.

Forecasting: Holt-Winters Exponential Smoothing (multiplicative trend + seasonality) on 117 months of data, validated with a 6-month hold-out (MAPE 6.6%).

Outlier detection: Z-score method flagged FY2018-19 and FY2023-24 as statistically unusual years within UPI's mature era (chosen over ML given only 9 annual data points).

Key Findings

Transaction value grew from Rs. 6,952 Cr (FY16-17) to over Rs. 26L Cr (FY24-25) — a structural shift in how India pays.

Fraud ratio's FY2023-24 spike is partly a reporting artifact: RBI reclassified 122 prior-year cases (Rs. 18,336 Cr) into FY2024-25 filings.

6-month forecast projects transaction value reaching ~Rs. 33.2L Cr by Feb 2027, continuing the current growth trajectory.

Limitation: RBI does not publicly release UPI-specific or state/channel-wise fraud data — Card/Internet category used as the closest official proxy.

Python (pandas, statsmodels)

SQL (PostgreSQL, window functions)

Power BI (DAX)

Holt-Winters Forecasting

Statistical Outlier Detection

[Your Name] | [your.email@example.com] | [LinkedIn URL] | [GitHub URL]

Built with Python, PostgreSQL, and Power BI. Full dashboard, SQL queries, and forecasting script available on GitHub. Data current as of the source publications' latest release (September 2026).